UK Price Paid — cleaned data handoff

Give teammates this folder (`data/`) plus this note. **Do not edit raw CSVs at repo root** (`pp-complete.csv`, monthly `* data.csv`); everything below is already derived.

Source: HM Land Registry Price Paid (England & Wales).

Pipeline (if regenerating): `scripts/build_uk_panel.py` → `scripts/analyze_trends_kmeans.py` → `scripts/analyze_rule_mining.py`.

What to use for what

| Goal | Use these files |

|---|---|

| Trend charts / EDA | `processed/uk_monthly_by_type.csv`, `processed/uk_county_year_stats.csv` |

| Cluster plots / cluster meaning | `processed/uk_sample_with_clusters.csv`, `processed/kmeans_cluster_centers.csv` |

| Association rules (Apriori) | `processed/association_rules.csv`, `processed/association_itemsets.csv`, `processed/association_rules_top25.csv` |

| Undo z-score → original scale | `processed/scaler_params_2018plus.csv` |

| Choose k (elbow / silhouette) | `processed/kmeans_k_metrics.csv` |

| Parallel k-means (Spark / MR / Python) | `parallel/uk_features_scaled_2018plus_sample10pct.txt` + `parallel/init_centroids_k4.txt` |

| Full 2018+ feature matrix (large) | `parallel/uk_features_scaled_2018plus.txt` (~572 MB) |

Inventory summary: `processed/data_inventory.txt` (or `.json`).

Property type codes

| Code | Meaning | `type_code` (numeric) |

|---|---|---|

| D | Detached | 0 |

| S | Semi-detached | 1 |

| T | Terraced | 2 |

| F | Flat / maisonette | 3 |

| O | Other | 4 |

File schemas

`processed/uk_monthly_by_type.csv`

One row per month × property type.

| Column | Meaning |

|---|---|

| `ym` | Year-month `YYYY-MM` |

| `property_type` | `D` / `S` / `T` / `F` / `O` |

| `n` | Transaction count |

| `median_price` | Median £ |

| `mean_price` | Mean £ |

| `mean_log_price` | Mean of log(price) |

Example (Python):

```
import pandas as pd
df = pd.read_csv("data/processed/uk_monthly_by_type.csv")
df["ym"] = pd.to_datetime(df["ym"] + "-01")
# median price over time for Detached
df[df["property_type"] == "D"].plot(x="ym", y="median_price")
```

`processed/uk_county_year_stats.csv`

One row per year × county/district.

| Column | Meaning |

|--------|---------|

| `year` | Calendar year |

| `county` | County / district name |

| `n` | Count |

| `median_price` | Median £ |

| `mean_log_price` | Mean log(price) |

| `new_share` | Share new-build (`Y`) |

| `freehold_share` | Share freehold |

`processed/uk_sample_with_clusters.csv` (and `_k4`)

Random sample of **z-scored** 2018+ features + cluster id (`k=4`).

| Column | Meaning |

|--------|---------|

| `log_price`, `year`, `month`, `type_code`, `is_new`, `is_freehold` | **Already standardised** (z-score) |

| `cluster` | 0–3 |

Cluster meaning (from centres):

| Cluster | Interpretation |

|---------|----------------|

| 0 | Freehold, early-year sales |

| 1 | Freehold, late-year sales |

| 2 | New-build homes |

| 3 | Leasehold / flats (lower price) |

To plot in **original** units, inverse-transform with `scaler_params_2018plus.csv`:

```
import pandas as pd
import numpy as np

sample = pd.read_csv("data/processed/uk_sample_with_clusters.csv")
params = pd.read_csv("data/processed/scaler_params_2018plus.csv").set_index("feature")
feats = ["log_price", "year", "month", "type_code", "is_new", "is_freehold"]
```

```
raw = sample[feats].to_numpy() * params.loc[feats, "scale"].to_numpy() + params.loc[feats, "mean"].to_nu
# raw[:, 0] = log(price), raw[:, 1] ≈ year, etc.
```

`processed/kmeans_cluster_centers.csv`

Cluster centres in the **same z-scored space** as the sample (columns = features).

`processed/scaler_params_2018plus.csv`

| Column | Meaning |

|--------|---------|

| `feature` | Feature name |

| `mean` | Train mean used for z-score |

| `scale` | Train std (ddof consistent with builder) |

`z = (x - mean) / scale` → `x = z * scale + mean`.

Parallel feature files (`data/parallel/`)

Whitespace-separated text, **6 columns, no header**, z-scored:

```
log_price  year  month  type_code  is_new  is_freehold
```

| File | Rows (approx.) |

|------|----------------|

| `uk_features_scaled_2018plus_sample10pct.txt` | 855,193 |

| `uk_features_scaled_2018plus.txt` | 8,551,939 |

| `init_centroids_k4.txt` | 4 lines (shared k-means init) |

Example:

```
import numpy as np
X = np.loadtxt("data/parallel/uk_features_scaled_2018plus_sample10pct.txt")
C0 = np.loadtxt("data/parallel/init_centroids_k4.txt")
# X.shape == (n, 6), C0.shape == (4, 6)
```

VM / Spark / MapReduce scripts live under `parallel/` and expect these paths (or the shared-folder mount of this repo).

Suggested zip for teammates

Small pack (EDA + clusters, ~15–30 MB):

```
data/processed/uk_monthly_by_type.csv
data/processed/uk_county_year_stats.csv
data/processed/uk_sample_with_clusters.csv
data/processed/kmeans_cluster_centers.csv
data/processed/scaler_params_2018plus.csv
data/processed/kmeans_k_metrics.csv
data/processed/data_inventory.txt
data/DATA_HANDOFF.md
```

Parallel pack (add ~58 MB):

```
data/parallel/uk_features_scaled_2018plus_sample10pct.txt
data/parallel/init_centroids_k4.txt
```

Skip sharing: root raw downloads, `uk_features_scaled_scale_n*.txt` (replication stress tests), log files under `processed/*nohup*`.

Figures

Report-ready plots (optional to share): `figures4papers/` + `figures4papers/CAPTIONS.md`.

Regenerate with: `python scripts/make_figures4papers.py`.